

IQgig-5G Model B: What You Need to Understand

A concise guide to the instrument architecture, TCP/SCPI interface, Python control pattern, and reliable measurement sequence.

1. The tester is three systems working together

- **VSG (Vector Signal Generator):** Generates and transmits the configured RF waveform.
- **VSA (Vector Signal Analyzer):** Captures RF and calculates EVM, power, occupied bandwidth, ACLR, spectrum emission mask, carrier leakage, and in-band emissions.
- **ROUT (Routing):** Connects physical RF ports to VSG and VSA resources. Incorrect routing prevents a valid test.

Normal measurement path: waveform generation -> RF routing -> signal capture -> 5G demodulation -> calculation -> result fetch

2. Important operating limits

Item	Capability
RF frequency range	23 GHz to 45 GHz
Maximum capture/modulation bandwidth	1.7 GHz
VSA maximum input	+20 dBm CW
VSG output range	Typically +10 dBm to -70 dBm below 40 GHz
RF ports	Two or four bidirectional 2.4 mm ports
Warm-up	60 minutes before specifications are valid
Default network address	192.168.100.254

Reference level, path-loss correction, cable condition, port routing, calibration state, and warm-up time can all change the apparent result. Record them with every diagnostic example.

3. Python controls the tester by sending SCPI over TCP

Python does not replace SCPI. It opens a TCP connection and sends the same SCPI program messages that an engineer would enter manually.

Port	Purpose
24000	Command/control: normal commands and queries
24001	Service request: asynchronous status notification
24002	Device clear: independent instrument clear

For ordinary automation, start with port 24000. SCPI uses ASCII text. Each program message and response ends with a line feed. A query ends in **?**, such as ***IDN?**.

4. Minimal Python connection

```
import socket

def query(sock, command):
    sock.sendall((command + "\n").encode("ascii"))
    response = b""
    while not response.endswith(b"\n"):
        chunk = sock.recv(65536)
        if not chunk:
            raise ConnectionError("Tester closed the connection")
        response += chunk
    return response.decode("ascii").strip()

with socket.create_connection(("192.168.100.254", 24000), timeout=10) as tester:
    print(query(tester, "*IDN?"))
```

Only read after sending a query. A setting command may return nothing, so an unconditional read can block until timeout.

5. The SCPI test sequence

- 1 Select the channel and 5G technology module.
- 2 Route physical RF ports to VSG and VSA resources.
- 3 Load and start the VSG waveform.
- 4 Set generator power.
- 5 Configure VSA capture frequency, time, sampling rate, reference level, and trigger.
- 6 Reset and configure 5G analysis: FR1/FR2, direction, bandwidth, modulation, resource blocks, DMRS, MIMO, and synchronization.
- 7 Start the VSA capture.
- 8 Run calculations such as transmit quality, power, spectrum, or CCDF.
- 9 Fetch the calculated values with the matching FETCh...? queries.

CALC and FETCh are different. CALC asks the tester to compute a measurement. FETCh returns it. The supplied command example calculates results but does not show the complete result-fetching layer required by Python.

6. Reliability rules for the Python layer

- Start with ***IDN?** and verify the expected tester.
- Use ***OPC?** or status handling to synchronize long captures and calculations.
- Use separate timeouts for configuration and measurement operations.
- After failure, read **SYST:ERR:ALL?** and preserve the full response.
- Interpret fetch status separately from the numeric value; a number is not automatically valid.
- Log commands, responses, timestamps, timeouts, settings, and software revision.
- Avoid competing controllers. External SCPI execution locks the browser GUI.
- Use the dashboard Trace Tool with source EXT to compare Python with known-good GUI commands.
- Turn off VSG output and close the socket in cleanup logic, including after exceptions.

7. Files that matter most

- **IQgig5GBUserGuideMaster.pdf:** TCP ports, framing, program messages, queries, and responses.
- **IQgig5GBQuickStart.pdf:** networking, GUI locking, Trace Tool, SCPI console, logs, and diagnostics.
- **IQgig5GBTechnicalSpecs.pdf:** RF limits, sampling rates, warm-up, and measurement capability.
- **module_5G.html:** exact 5G configuration, calculation, and fetch commands.
- **module_VSA.html, module_VSG.html, module_ROUT.html:** analyzer, generator, and routing commands.
- **FetchStatusCodes.html and errors.html:** result validity and failure interpretation.
- **SCPI_commanda.txt:** the existing manual sequence and starting point for automation.

Sources

LitePoint IQgig-5G Model B User Guide; Quick Start Guide; Technical Specifications; and SCPI HTML reference, revision 1.24.0.288143.